

# AAKASH KUMAR AGARWAL

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Machine Learning Engineer and published AAAI/EMNLP researcher (MS, IIT Bombay) specializing in production LLM systems, post-training pipelines, and agentic AI workflows. 4+ years combined experience across applied research and software engineering, with deep hands-on work in PyTorch, Hugging Face Transformers, LoRA/PEFT, DPO/RLHF, RAG, vector retrieval, and LangChain/LangGraph multi-agent systems. Built reasoning frameworks lifting LLM accuracy by 56.7%, large-scale preference-optimization pipelines on 150K+ pairs, and evaluation harnesses spanning GPT-4o, Qwen2.5, Gemma-3, and LLaMA-3. Strong systems background in distributed training, inference optimization, Docker/MLflow-based experimentation infrastructure, and shipping reliable software.

## EDUCATION

**Indian Institute of Technology (IIT) Bombay**, Mumbai, India | CGPA 8.83 Aug 2023 – Jun 2026 (Expected)  
*M.S. by Research, Computer Science & Engineering (AI/ML)*

**Amity University**, Noida, India | CGPA 7.65 Aug 2015 – Jun 2019  
*B.Tech, Computer Science & Engineering*

## TECHNICAL SKILLS

**Languages:** Python, C++, SQL, Shell/Bash, HTML/CSS, **MLOps & Infra:** Docker, MLflow, Git, FastAPI

**ML / Deep Learning:** PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, NumPy, Pandas, Matplotlib

**LLMs & Post-Training:** LoRA, QLoRA, PEFT, DPO, GRPO, RLHF, SFT, Fine-tuning, Hypernetworks, Distributed Training, DeepSpeed, Accelerate

**LLM Systems & Agents:** LangChain, LangGraph, RAG, Vector Databases (FAISS, Chroma), Multi-Agent Systems, Agentic AI, Prompt Engineering, vLLM, Inference Optimization

**Research Domains:** NLP, LLM Reasoning, LLM Unlearning, Alignment, Generative Models (Diffusion), Retrieval Systems

## EXPERIENCE

**Fujitsu Research** — AI Engineer Intern - *Research output: AAAI 2026 (Main Track)* May 2025 – Jul 2025

- Architected a modular Hypothesis-Driven Reasoning (HDR) framework for LLM inference, decoupling hypothesis generation, verification, and refinement into composable Python services; benchmarked across GPT-4o, Qwen2.5, and Gemma-3, improving reasoning-task accuracy by up to 56.7 percentage points on multi-step experience-transfer benchmarks.
- Engineered a two-stage long-term memory architecture for LangChain/LangGraph agents with FAISS-backed vector retrieval, cutting redundant LLM API calls by 43% and reducing average inference latency per reasoning episode from 18s to 10s.
- Built a YAML-driven, MLflow-tracked evaluation harness across 3 model families and 12 reasoning benchmarks (2,400+ runs), cutting experiment turnaround time from ~2 days to ~6 hours and enabling 3x faster ablation cycles.
- Containerized full experimentation stack (Docker + Hugging Face Transformers) on 4-GPU internal cluster; reduced environment setup time from ~3 hours to under 10 minutes for new team members; co-authored AAAI 2026 Main Track paper.

**CFILT Lab, IIT Bombay** — Student AI Engineer - *Research output: EMNLP 2025 (Main Track)* Jan 2024 – Dec 2025

- Designed end-to-end NLP pipelines on large-scale social-media corpora (~1.2M posts) using transformer-based encoders (BERT, RoBERTa, LLaMA-3), shipping a depression-relapse prediction system that achieved 0.88 accuracy and 0.84 F1 on expert-annotated benchmarks — outperforming prior SOTA by 9.3 F1 points — accepted at EMNLP 2025.
- Built data ingestion, preprocessing, and feature-engineering pipelines in PyTorch and Hugging Face, reducing raw-to-training-ready data processing time by 58%; containerized training with Docker across 3 GPU nodes for reproducible cluster runs tracked via MLflow.

- Developed multilingual robustness and safety evaluation workflows covering 6 languages, 400+ adversarial probes, and cognitive-load stress tests — identifying a degradation in safety robustness in mid-size LLMs on multilingual inputs.
- Leading HyperLoRA: a hypernetwork-driven framework generating LoRA adapter weights for parameter-efficient LLM unlearning; achieved 97.2% forget-set erasure while retaining 94.1% general benchmark performance on TOFU and SISA evaluation suites.

**Accenture** — Software Engineer

May 2019 – Dec 2021

- Delivered 14+ backend features across 3 enterprise applications using Python, REST APIs, and SQL, contributing to implementation, testing, deployment, and production support within 2-week agile sprints.
- Maintained Python backend services processing 500K+ daily requests, improving average API response times by 22% through query optimization and Redis caching while resolving 120+ production issues with high SLA adherence.

## SELECTED PUBLICATIONS

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- [Hypothesis-Driven Reasoning for Large Language Models](#) — A. K. Agarwal\*, M. Yamada\*. *AAAI 2026 (Main Track, accepted)*. Proposes an iterative hypothesis-generation/verification framework for LLM reasoning with reusable memory.
- [ReDepress: A Cognitive Framework for Predicting Depression Relapse from Social Media](#) — A. K. Agarwal\*, S. Bhattacharjee\*, P. Bhattacharyya. *EMNLP 2025 (Main, accepted)*. Transformer-based temporal modeling of cognitive-bias signals for clinical-grade relapse prediction.

## SELECTED PROJECTS

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**HyperLoRA — Hypernetwork-Driven LLM Unlearning** | *PyTorch, Hugging Face, LoRA, PEFT, Hypernetworks, TOFU/SISA Benchmarks*

- Designed a parameter-efficient unlearning framework where a learned hypernetwork generates request-specific LoRA adapter weights on the fly, enabling scalable targeted forgetting without retraining base model parameters.
- Implemented end-to-end training, unlearning, and evaluation pipelines on TOFU and SISA benchmarks; instrumented evaluation across forget-set accuracy, retain-set utility, and downstream reasoning preservation.

**Profession-Aware LLM Personalization via Preference Optimization** | *Gemma-2-2B, LoRA, DPO, GRPO, Hugging Face TRL, BERT Reward Models*

- Built an end-to-end preference-optimization pipeline over 150K preference pairs, fine-tuning Gemma-2-2B with LoRA + DPO + GRPO for controllable, profession-conditioned response adaptation.
- Developed an automated evaluation framework spanning 8 behavioral dimensions using BERT-based reward models, enabling reproducible regression testing across model checkpoints.

**GradSafe — Gradient-Based Jailbreak Detection for LLMs** | *PyTorch, Adversarial ML, Safety-Critical Parameter Analysis*

- Built a runtime jailbreak detector that analyzes gradients on safety-critical parameters to flag adversarial prompts, achieving >70% F1 on standard jailbreak benchmarks while remaining model-agnostic across open LLM families.

**Multimodal Emotion Recognition (Speech + Text)** | *LLaMA-2, Whisper, LoRA, PyTorch*

- Adapted a LLaMA-2-based architecture with Whisper speech embeddings using LoRA fine-tuning for multimodal emotion classification, fusing acoustic and textual features into a single conditioned representation.

**Neural Hashing for Large-Scale Image Retrieval** | *Vision Transformers (ViT), Learned LSH, PyTorch*

- Implemented a ViT-based image-retrieval system with learned hash codes (neural LSH), achieving 96% mAP on CIFAR-10 and demonstrating sub-linear retrieval over large embedding stores.

**Other Open-Source Work** | *GitHub: [github.com/aakash-agarwal-002](https://github.com/aakash-agarwal-002)*

- [Supercharge-YouTube, JioHotstar and SonyLiv](#) (published Chrome extension).

## ACHIEVEMENTS & HIGHLIGHTS

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- **GATE CS 2023: All India Rank 156** out of 75,680 candidates (top 0.2 percentile) — admission to IIT Bombay MS
- **Two top-tier publications** in consecutive cycles: **AAAI 2026** (Main Track) and **EMNLP 2025** (Main Track) — both as first/co-first authors.